

Lottery Scheduling

Probabilistic CPU sharing with tickets

Definition

Lottery scheduling is a scheduling algorithm that gives processes lottery tickets for a resource. Whenever the resource must be allocated, the scheduler randomly chooses one ticket, and the process holding that ticket gets the resource.

1. Motivation

Problem with Strict Guarantees

Guaranteed scheduling gives predictable fairness, but it requires careful accounting and can be difficult to implement exactly.

Lottery Idea

Lottery scheduling gives similar long-run fairness with a simpler mechanism: assign tickets, draw a random winning ticket, and run the process that owns it.

Historical Note

Lottery scheduling was introduced by Waldspurger and Weihl in 1994 as a randomized way to manage resource allocation.

2. Basic CPU Scheduling Rule

Scheduling Lottery

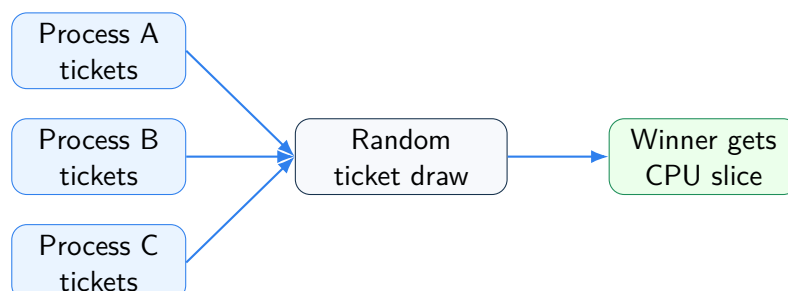
When the CPU becomes available, the operating system holds a lottery among all runnable processes. The process holding the winning ticket receives the next CPU time slice.

Time-Slice Example

The system might hold a lottery 50 times per second. Each winner then receives:

$$\frac{1}{50} \text{ second} = 20 \text{ ms}$$

of CPU time.



3. Tickets Represent Shares

Ticket Fraction

If a process owns a fraction f of all outstanding tickets, then it has probability f of winning each lottery. In the long run, it should receive about fraction f of the resource.

$$f = \frac{\text{tickets held by process}}{\text{total tickets in system}}$$

Clear Meaning

Unlike a priority number, a ticket count has a direct interpretation. If a process owns 20% of the tickets, it should receive about 20% of the CPU over time.

4. Simple Ticket Example

100 Total Tickets

Suppose there are 100 tickets total, and process A owns 20 of them. Then:

$$\frac{20}{100} = 0.20 = 20\%$$

Process A has a 20% chance of winning each lottery and should receive about 20% of the CPU in the long run.

Process	Tickets	Chance per Lottery	Expected CPU Share
A	20	$20/100 = 20\%$	About 20%
B	30	$30/100 = 30\%$	About 30%
C	50	$50/100 = 50\%$	About 50%



100 total tickets

5. Unequal Importance

More Tickets, More Chance

All processes can participate in the lottery, but important processes can be given more tickets. More tickets mean a higher probability of winning CPU time.

Main Rule

If one process should receive twice as much CPU as another, give it twice as many tickets.

Short Run vs. Long Run

Lottery scheduling is probabilistic. A process with 20% of the tickets does not win exactly every fifth lottery. It receives about 20% of the CPU over a longer period.

6. Algorithm

```
total_tickets = sum tickets of all runnable processes
winning_ticket = random number from 1 to total_tickets

current_sum = 0
for each runnable process:
    current_sum = current_sum + process.tickets
    if current_sum >= winning_ticket:
        run this process for one time slice
        break
```

Implementation Point

The scheduler only needs ticket counts and a random draw. It does not need to compute exact future CPU usage for every process.

7. Responsiveness

New Process

If a new process arrives and receives tickets, it can win immediately at the next lottery.

Why It Is Responsive

The new process does not need to wait for a long accounting adjustment. Its chance of running appears as soon as its tickets are included in the lottery.

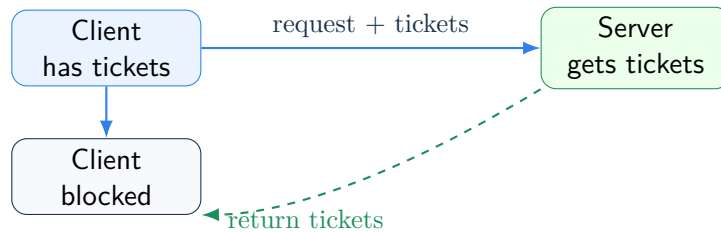
8. Ticket Transfer

Ticket Transfer

Cooperating processes can temporarily transfer tickets to each other. This lets CPU share move to the process that is currently doing useful work.

Client and Server

When a client sends a request to a server and then blocks, the client can give its tickets to the server. The server is then more likely to run and finish the request quickly. When the server finishes, it returns the tickets to the client.



Server Without Clients

If no clients need service, a server may need no tickets at all. Tickets can be supplied by clients only while the server is doing work for them.

9. Video Server Example

Different Frame Rates

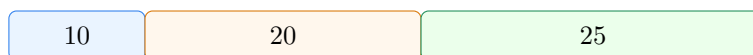
Suppose three video-stream processes need frames at 10, 20, and 25 frames per second. Give them 10, 20, and 25 tickets.

10 : 20 : 25

Stream Process	Required Frame Rate	Tickets	CPU Share
P1	10 frames/sec	10	$10/55 \approx 18.2\%$
P2	20 frames/sec	20	$20/55 \approx 36.4\%$
P3	25 frames/sec	25	$25/55 \approx 45.5\%$

Result

The CPU is divided automatically in approximately the required ratio 10 : 20 : 25. Processes needing higher frame rates receive more tickets and therefore win more often.



ticket ratio matches frame-rate ratio

10. Comparison with Priority Scheduling

Approach	Meaning of Control Value	Result
Priority scheduling	A priority number says which process is more important, but the exact CPU share may be unclear.	Good for ordering, less direct for proportional sharing.
Lottery scheduling	Ticket fraction directly represents expected resource fraction.	Clear proportional sharing in the long run.

Clear Rule

A process holding fraction f of the tickets gets about fraction f of the resource.

11. Strengths and Weaknesses

Aspect	Explanation
Main benefit	Gives predictable long-run proportional sharing with a simple randomized algorithm.
Responsive behavior	New processes can win at the next lottery after receiving tickets.
Flexible control	More important processes can receive more tickets.
Cooperation support	Processes can transfer tickets to speed up related work.
Main limitation	Short-term results are random, so exact fairness is only approximate over time.

12. Big Picture

Main Conclusion

Lottery scheduling allocates CPU time by randomly choosing among tickets. The more tickets a process holds, the more often it should win. This gives simple, responsive, proportional sharing without the detailed accounting required by strict guaranteed scheduling.

Exam Shortcut

Remember: tickets are probabilities. If a process owns 20 out of 100 tickets, it has a 20% chance of winning each lottery and should receive about 20% of the CPU in the long run.

13. Quick Review

Key Points

- Lottery scheduling gives processes tickets for resources such as CPU time.
- Each scheduling decision is made by drawing a random winning ticket.
- The process holding the winning ticket receives the next CPU time slice.
- If lotteries happen 50 times per second, each winner receives about 20 ms.
- A process with fraction f of the tickets receives about fraction f of the resource in the long run.
- More important processes can be given more tickets.
- Lottery scheduling is responsive because new ticket holders can win immediately.
- Cooperating processes can transfer tickets.
- Clients can give tickets to servers while waiting for replies.
- A video server can allocate tickets in proportion to frame-rate needs, such as 10 : 20 : 25.